WRITE A TERRAFORM SCRIPT TO DEPLOY EKS CLUSTER

1.main.tf

```
#############################
# Default VPC & Subnets
#############################

data "aws_vpc" "default" {
  default = true
}

data "aws_subnets" "default" {
  filter {
    name   = "vpc-id"
    values = [data.aws_vpc.default.id]
  }
}

#############################
# EKS Cluster IAM Role
#############################

resource "aws_iam_role" "eks_cluster_role" {
  name = "eks-cluster-role"

  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Effect = "Allow"
        Principal = {
          Service = "eks.amazonaws.com"
        }
        Action = "sts:AssumeRole"
      }
    ]
  })
}
```

```
resource "aws_iam_role_policy_attachment" "eks_cluster_policy" {
  role       = aws_iam_role.eks_cluster_role.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEKSClusterPolicy"
}

##############################
# EKS Cluster
##############################

resource "aws_eks_cluster" "eks" {
  name     = "my-eks-cluster"
  role_arn = aws_iam_role.eks_cluster_role.arn

  vpc_config {
    subnet_ids = data.aws_subnets.default.ids
  }

  depends_on = [
    aws_iam_role_policy_attachment.eks_cluster_policy
  ]
}

##############################
# Node Group IAM Role
##############################

resource "aws_iam_role" "eks_node_role" {
  name = "eks-node-group-role"

  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Effect = "Allow"
        Principal = {
          Service = "ec2.amazonaws.com"
        }
        Action = "sts:AssumeRole"
      }
    ]
```

```
  })
}

resource "aws_iam_role_policy_attachment" "worker_node_policy" {
  role       = aws_iam_role.eks_node_role.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEKSWorkerNodePolicy"
}

resource "aws_iam_role_policy_attachment" "cni_policy" {
  role       = aws_iam_role.eks_node_role.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEKS_CNI_Policy"
}

resource "aws_iam_role_policy_attachment" "ecr_readonly" {
  role       = aws_iam_role.eks_node_role.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEC2ContainerRegistryReadOnly"
}

##############################
# EKS Node Group
##############################

resource "aws_eks_node_group" "nodes" {
  cluster_name    = aws_eks_cluster.eks.name
  node_group_name = "worker-nodes"
  node_role_arn   = aws_iam_role.eks_node_role.arn

  subnet_ids = data.aws_subnets.default.ids

  instance_types = ["c7i-flex.large"]

  scaling_config {
    desired_size = 2
    min_size     = 1
    max_size     = 3
  }

  depends_on = [
    aws_iam_role_policy_attachment.worker_node_policy,
    aws_iam_role_policy_attachment.cni_policy,
```

```
    aws_iam_role_policy_attachment.ecr_readonly
  ]
}
```

2.output.tf

```
output "cluster_name" {
  value = aws_eks_cluster.eks.name
}
output "cluster_endpoint" {
  value = aws_eks_cluster.eks.endpoint
}
output "node_group_name" {
  value = aws_eks_node_group.nodes.node_group_name
}
```

3.provider.tf

```
provider "aws" {
    region = "ap-south-1"
    profile = "config"
}
```

1. git add .
2. git commit -m “eks-cluster”
3. git push

ON UBUNTU-LOCAL

1. git clone https://github.com/mayurpandit25/terraform-b30.git
2. cd terraform-b30
3. terraform init
4. terraform apply –auto-approve
5. terraform destroy –auto-approve
